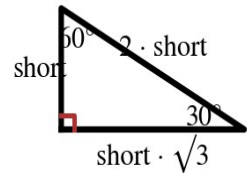
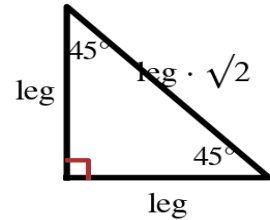


Speak Like A Mathematician:

Where do the two special right triangles come from?



Special Right Triangle Relationships

45°-45°-90°: the hypotenuse is _____ times as long as each leg (and the two legs are congruent).

30°-60°-90°: the hypotenuse is _____ times the SHORTER leg, and the longer leg is _____ times the shorter leg.

The shorter leg is always opposite the _____° angle; the longer leg is opposite the _____° angle.

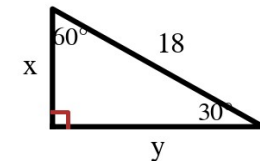
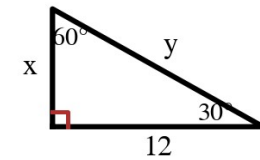
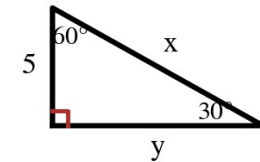
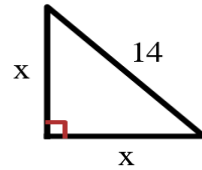
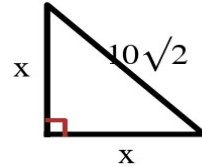
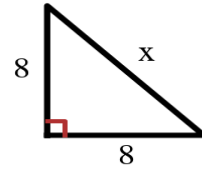
Example 1 - Rationalize Each Denominator (Algebra 1 recall)

- a) $5/\sqrt{3}$ and $7/\sqrt{2}$
- b) $4/\sqrt{8}$ and $6/\sqrt{12}$
- c) $\sqrt{3}/\sqrt{6}$

Example 2 - 45°-45°-90°

Give every answer in simplest radical form. Figures at the right.

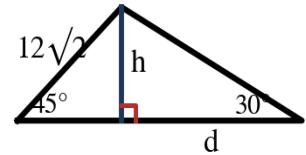
- a) Legs 8. Find the hypotenuse x .
- b) Hypotenuse $10\sqrt{2}$. Find each leg x .
- c) Hypotenuse 14. Find each leg x .



Example 4 - A Composite Figure

In the figure at the right, the altitude h splits the big triangle into a 45° - 45° - 90° triangle (left) and a 30° - 60° - 90° triangle (right). The left hypotenuse is $12\sqrt{2}$.

- a) Find h .
- b) Find d .



Example 5 - Special Triangles Hiding in Shapes

- a) A square has diagonal 20. Find its side length (exact).
- b) An equilateral triangle has side 12. Find its altitude (exact).
- c) Why do these two shapes ALWAYS hide special right triangles?